INCH-POUND

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SUPERSEDING
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PERFORMANCE SPECIFICATION

CONTROL, DIRECTIONAL SIGNAL LIGHT, AUTOMOTIVE: 24 VOLT DC, SOLID STATE

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers control and flasher units of the solid state type used on military vehicles.

1.2 Classification. For the purpose of this specification, components are classified as follows (see 6.2):

Unit 1	- Control Unit
Unit 2	- Flasher Unit

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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2.2 Government documents

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50588	- Lamp, T-3-1/4, Miniature Bayonet, Single Contact.
A-A-52463	- Lamp Bulbs, Incandescent, Miniature, Single and Double Contact, Bayonet Candelabra Base.

DEPARTMENT OF DEFENSE

MIL-C-13516/1	- Circuit Breakers; 28 Volt DC, Automatic, Waterproof, 15 to 30 Amp.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-193	- Painting Procedures and Marking for Vehicles, Construction Equipment, Material Handling Equipment, and Spare Parts.
MIL-STD-202	- Electronic and Electrical Component Parts, Test Methods for.
MIL-STD-461	- Control of Electromagnetic Interference Emissions and Susceptibility, Requirements for the.
MIL-STD-462	- Measurement of Electromagnetic Interference Characteristics.
MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Tests.
MS 35000	- Battery, Storage, Lead-Acid, Waterproof.

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS

ARMY

11613631	- Flasher, Solid State.
11613632	- Control Assy. Directional Signal (Waterproof).

(Copies of the drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

DEPARTMENT OF TRANSPORTATION (DOT)

Code of Federal Regulations, Title 10, Part 40.
Code of Federal Regulations, Part 49.

(Applications for copies should be addressed to the U.S. Department of Transportation, Superintendent of Documents, Government Printing Office, Washington, DC 20402).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4	- Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Application for copies should be addressed to American National Standards Institute, 11 West 42nd Street, New York, NY 10036).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1	- Unified Inch Screw Threads (UN and UNR Thread Form) (DoD Adopted).

(Application for copies should be addressed to American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE AS 478 - Identification Marking Methods.

(Application for copies should be addressed to Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials shall be as specified herein, on applicable specifications, and on applicable drawings. Materials not specifically designated shall be suitable for use in the control and flasher units operating over an ambient temperature range from -65° to 160°F without any change in physical or dimensional properties that would result in operation of the units falling outside of the limits specified herein. Asbestos, cadmium, and radioactive material shall not be used in this item. Radioactive material is defined by, Code of Federal Regulations,Title 10, Part 40, and as other radioactive material in which the radioactivity is greater than 0.002 microcurries per gram or 0.01 microcurries total activity for the item.

3.2.1 Potting compound. Potting compound, when used, shall be non-hygroscopic, shall comply with shock and vibration requirements specified herein, and shall not support fungus growth.

3.2.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design and construction. The control and flasher covered by this specification shall conform to the interface, volumetric, and mounting requirements of Drawings 11613632 and 11613631, respectively. This signal light control shall comply with all applicable Department of Transportation requirements found in part 49 of the Code of Federal Regulations.

3.3.1 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals which would encourage galvanic action shall be avoided.

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3.3.2 Standard parts. Military standard parts shall be used wherever possible, if repair parts are called out for the control and flasher.

3.3.3 Threaded parts. All screw threads shall conform to ASME B1.1.

3.3.4 Flash indicator lamp. The flash indicator lamp for the control unit shall conform to A-A-50588. The lamp indicator light opening shall be located as shown on drawing 11613632. Luminescence from the lamp opening shall be a green color without glare that would blind the driver. Emitting surface shall have a minimum area equivalent to a 0.19 inch diameter circle.

3.3.4.1 Flash indicator lamp replacement. Replacement of the flash indicator lamp for the control unit shall be capable of being accomplished without removing the control unit from the steering column. Access for replacement shall be accomplished by removing the lens or by loosening the captive retaining devices permitting movement of a captive subassembly of the control unit. All components exposed during replacement of the bulb shall be captive (including seals). When the lens assembly or the captive subassembly of the control unit is located on the basic control unit housing, the operation of replacing the lamp shall not expose the switch contact or any other moving internal parts of the control unit to the entry of foreign matter, and the unit shall maintain the waterproofness requirement after lamp replacement.

3.3.5 Hazard warning switch. The control unit shall have a hazard warning switch in a location such that accidental operation shall be prevented. The hazard warning switching device shall disengage when the signal control lever is actuated.

3.4 Electrical.

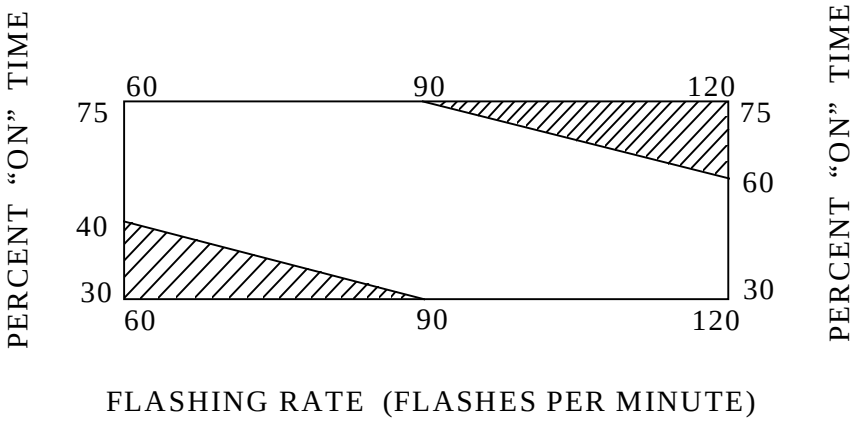
3.4.1 Voltage rating. Control and flasher units shall be suitable for use in nominal 24 volt direct current (dc) electrical systems.

3.4.2 Current rating. Except as otherwise indicated, current carrying parts shall be able to withstand a continuous dc current of 10 amperes without damage.

3.4.3 Load impedance. The flasher unit shall be operable with an inductive load of 0.20 millihenries placed in series with the lamp load.

3.4.4 Flashing. When tested as specified in 4.5.2.1, the “flashing rate” and the “percent on time” of normally closed type flashers shall be within the unshaded area of the polygon below.

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"Flashing rate" and "percent on time" shall be measured after the flasher has been operating for five consecutive cycles and shall be an average of at least three consecutive cycles. The "flashing rate", as defined in 6.4.4, shall be not less than 60 nor more than 120 flashes per minute. The average "percent on time", as defined in 6.4.5, shall be as indicated in the polygon above. When the control lever or hazard warning switch is returned to neutral position, the "turn off time", as defined in 6.4.7, shall be less than 1.0 second.

3.4.5 Overload.

3.4.5.1 Control unit. When tested as specified in 4.5.2.2.1, the control unit shall not be adversely affected by the application of a simulated short circuit condition.

3.4.5.2 Flasher unit. When tested as specified in 4.5.2.2.2, the flasher unit shall not be adversely affected by the application of a simulated short circuit condition. The flasher unit shall be short-circuit protected internally and shall not contain a fuse or circuit breaker. Subsequent to the test, the flasher shall meet the requirements of 3.4.4.

3.4.6 Reverse polarity. After testing as specified in 4.5.2.3, the flasher unit shall meet the requirements of 3.4.4. The unit shall not be operational with reverse polarity.

3.4.7 Endurance. When tested as specified in 4.5.2.4, the control unit shall operate without malfunction and meet the requirements of 3.5.1 throughout the test. The flasher unit shall meet the requirements of 3.4.4 throughout the test.

3.4.8 Transient capability. After testing as specified in 4.5.2.5, the flasher unit shall meet the requirements of 3.4.4.

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3.4.9 Electromagnetic interference. The turn signal flasher and control unit shall not exceed allowable electromagnetic emissions as tested per contractor's Government approved plan (see 4.5.2.6).

3.5 Mechanical.

3.5.1 Actuation. When tested as specified in 4.5.3.1, the force required to move the directional control lever from any position to an adjacent position shall be not less than two nor more than four pounds. To move the hazard warning switch from any position to any adjacent position, the force shall be not less than two nor more than seven pounds.

3.5.2 Strength, control unit. The actuation lever shall withstand the test specified in 4.5.3.2, without any permanent distortion or other damage that would impair the functional operation of the control unit.

3.6 Environmental.

3.6.1 High temperature.

3.6.1.1 Control unit. When tested as specified in 4.5.4.1.1, the control unit shall operate in a temperature of 160° ± 5°F without malfunction. The control unit shall meet the requirements of 3.5.1 throughout the test.

3.6.1.2 Flasher unit. When tested as specified in 4.5.4.1.2, the flasher unit shall operate in a temperature of 160° ± 5°F without malfunction. The flasher unit shall meet the requirements of 3.4.4 throughout the test.

3.6.2 Low temperature.

3.6.2.1 Control unit. When tested as specified in 4.5.4.2.1, the control unit shall operate in a temperature of -65° ± 5°F without malfunction. The control unit shall meet the requirements of 3.5.1 throughout the test.

3.6.2.2 Flasher unit. When tested as specified in 4.5.4.2.2, the flasher unit shall operate in a temperature of -65° ± 5°F without malfunction. The flasher unit shall meet the requirements of 3.4.4 throughout the test.

3.6.3 Shock. When tested as specified in 4.5.4.3, the control and flasher units shall evidence no loosened, distorted, or broken parts. The signal control lever shall not be displaced during the test. Subsequently, the control unit shall meet the requirements of 3.5.1 and the flasher unit shall meet the requirements of 3.4.4.

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3.6.4 Vibration. When tested as specified in 4.5.4.4, the control and flasher units shall evidence no loosened, distorted, or broken parts. There shall be no oscillation of the control unit body about the cylinder axis, or displacement of the control lever. Subsequently, the control unit shall meet the requirements of 3.5.1 and the flasher unit shall meet the requirements of 3.4.4.

3.6.5 Waterproofness. When tested as specified in 4.5.4.5, the control and flasher units shall evidence no leakage or damage which would adversely affect performance. Subsequently, the control unit shall meet the requirements of 3.5.1 and flasher unit shall meet the requirements of 3.4.4.

3.6.6 Corrosion resistance. When tested as specified in 4.5.4.6, the control and flasher units shall show no evidence of corrosion that would adversely affect performance. Subsequently, the control unit shall meet the requirements of 3.5.1 and the flasher unit shall meet the requirements of 3.4.4.

3.7 Finish.

3.7.1 Appearance. The control unit shall have an olive drab finish and the flasher unit shall have an olive drab or black finish. Painting processes shall conform to applicable military standards (see 4.4.2.1).

3.8 Marking. The control and flasher units shall both be marked in accordance with SAE AS478 and each shall include as a minimum the following :

a. 24 volts DC
b. Manufacturer’s Identification
c. Manufacturer’s Part Number
d. Serial Number
e. Military Part Number
f. Federal Stock Number
g. US

3.9 Workmanship. Workmanship shall be of a quality which will assure a product free of burrs, rust, scratches, chips, sharp edges, loose or defective connectors, cracked insulation, faulty soldering or other defects which affect serviceability or appearance. The contractor shall provide for Government inspection of preliminary phases to assure correct processing.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. The first article sample (see 3.1) shall consist of four specimens each of the control and flasher unit. They shall be properly identified as representative of the units proposed to be delivered to the Government. First article sampling and testing is the responsibility of the manufacturer unless modified by the contract. Government surveillance of testing to assure examination for defects specified in 4.3.2.2 and testing as shown in table I and table II, as applicable, is required.

TABLE I. Order of first article testing for control unit.

Specimen	Paragraph No.	Tests
1 and 2	4.5.3.1	Actuation
	4.5.2.2.1	Overload
	4.5.3.1	Actuation
	4.5.2.4	Endurance (Control unit)
	4.5.3.1	Actuation
	4.5.4.1.1	High temperature
	4.5.3.1	Actuation
	4.5.4.2.1	Low temperature
	4.5.3.1	Actuation
	4.5.2.4	Endurance (Warning switch)
	4.5.3.1	Actuation
	4.5.3.2	Strength, control unit
	4.5.4.5	Waterproofness
	4.5.3.1	Actuation
3	4.5.3.1	Actuation
	4.5.2.2.1	Overload
	4.5.3.1	Actuation
	4.5.4.4	Vibration
	4.5.3.1	Actuation
	4.5.4.3	Shock
	4.5.3.1	Actuation
	4.5.3.2	Strength, control unit
	4.5.3.1	Actuation
4	4.5.3.1	Actuation
	4.5.4.6	Corrosion resistance
	4.5.3.1	Actuation

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TABLE II. Order of first article testing for flasher unit.

Specimen	Paragraph No.	Tests
1	4.5.2.1	Flashing
	4.5.4.2.2	Low temperature
	4.5.2.1	Flashing
	4.5.4.1.2	High temperature
	4.5.2.1	Flashing
	4.5.2.4	Endurance
	4.5.2.1	Flashing
2	4.5.2.1	Flashing
	4.5.2.5	Transient capability
	4.5.2.1	Flashing
	4.5.2.2.2	Overload
	4.5.2.1	Flashing
	4.5.2.3	Reverse polarity
	4.5.2.1	Flashing
	4.5.4.4	Vibration
	4.5.2.1	Flashing
	4.5.4.5	Waterproofness
	4.5.2.2.2	Overload
	4.5.2.1	Flashing
3	4.5.2.1	Flashing
	4.5.4.3	Shock
	4.5.2.1	Flashing
	4.5.4.6	Corrosion resistance
	4.5.2.1	Flashing
	4.5.2.2.2	Overload
	4.5.2.1	Flashing
4	4.5.2.6	Electromagnetic interference

4.2.1 Failure. Failure of any first article specimen to pass any specified examination or test may be cause, at the option of the Government, for refusal to conduct additional testing until the faults revealed by the tests have been corrected.

4.3 Conformance inspection.

4.3.1 Sampling.

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4.3.1.1 Lot formation. Unless otherwise specified (see 6.2), a lot shall consist of all units of the same kind from an identifiable production period from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Sampling for examination. Samples for conformance examination shall be selected in accordance with ANSI/ASQC Z1.4.

4.3.1.3 Sampling for testing. A representative sample shall be selected in accordance with applicable inspection level of ANSI/ASQC Z1.4 from each lot that has passed the examination specified in 4.3.2.2.

4.3.2 Examination.

4.3.2.1 Acceptability. Each specimen selected in accordance with 4.3.1.2 shall be examined for conformance with ANSI/ASQC Z1.4.

4.3.2.2 Classification of defects. For examination purposes, defects shall be classified as specified in table III.

TABLE III. Classification of defects for component units.

Categories	Defects	Method of examination
Major:		
101	Incorrect dimensions affecting interchangeability (see 3.3)	SIE 1 /
Minor:		
201	Incorrect dimensions not affecting interchangeability (see 3.3)	SIE
202	Incorrect luminescent area (control unit only) (see 3.3.4)	SIE
203	Improper finish (see 3.7)	Visual
204	Improper marking (see 3.8)	Visual
205	Faulty workmanship (see 3.9)	Visual

1 / SIE - Standard Inspection Equipment.

4.3.3 Conformance tests for flasher. Samples selected in accordance with 4.3.1.3 shall be subjected to the tests specified in table IV.

4.3.4 Conformance tests for control units. All (100%) control units shall be subjected to the tests specified in table IV.

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TABLE IV. Order of conformance tests.

Sample	Paragraph	Tests	
Flasher	4.5.3.1	Actuation	(See Note 1)
	4.5.2.1	Flashing	(See Note 2)
	4.5.2.2.2	Overload	(See Note 3)
	4.5.2.1	Flashing	(See Note 2)
	4.5.2.3	Reverse Polarity	(See Note 4)
	4.5.2.1	Flashing	(See Note 2)
Control Unit	4.5.3.1	Actuation	(See Note 1)
	4.5.2.1	Flashing	(See Note 2)
	4.5.4.5	Waterproofness	(See Note 5)
	4.5.2.1	Flashing	(See Note 2)
	4.5.3.1	Actuation	(See Note 1)

Note 1: Perform in accordance with 4.5.3.1 but actuate twice.
Note 2: Check each unit to 18, 24 and 30 volts. Eliminate "Average flashing rate" and "percent on time" specified in 3.4.4.
Note 3: Short for approximately 15 seconds.
Note 4: Apply reverse polarity only once.
Note 5: Submerge in tap water one minute. Electrical operation not required. A continuous stream of bubbles from the unit shall be cause for failure.

4.4 Control tests.

4.4.1 Sampling. Control test samples shall be selected at the rate of two for each 500 units produced, except that not more than six nor less than two units shall be selected in any given 30 day period. The control test sample shall be selected from a lot which has passed the examination specified in 4.3.2 and tests specified in table IV, and shall be subjected to the tests specified in table V in the order listed.

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TABLE V. Order of control tests.

Sample	Paragraph	Tests
Control unit 1	4.5.3.1	Actuation
	4.5.2.2.1	Overload
	4.5.3.1	Actuation
	4.5.4.5	Waterproofness
	4.5.3.1	Actuation
Control unit 2	4.5.3.1	Actuation
	4.5.2.2.1	Overload
	4.5.3.1	Actuation
	4.5.4.4	Vibration
	4.5.3.1	Actuation
	4.5.4.3	Shock
	4.5.3.1	Actuation
Flasher unit 1	4.5.2.1	Flashing
	4.5.2.5	Transient capability
	4.5.2.1	Flashing
	4.5.2.2.2	Overload
	4.5.2.1	Flashing
	4.5.2.3	Reverse polarity
	4.5.2.1	Flashing
Flasher unit 2	4.5.2.1	Flashing
	4.5.4.2.2	Low temperature
	4.5.2.1	Flashing
	4.5.4.1.2	High temperature
	4.5.2.1	Flashing

4.4.2 Failure. Failure of any control test sample to pass any specified examination or test may be cause, at the option of the Government, for refusal to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the test have been corrected.

4.4.2.1 Painting inspection. External painted surfaces processes shall be approved by the Government inspection to assure the cleaning, conditioning, priming and painting phases of finishing conform to MIL-STD-193 (see 3.9).

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4.5 Test procedures.

4.5.1 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified below (see 4.5.1.1, 4.5.1.2 and 4.5.1.3).

4.5.1.1 Conditions. Tests shall be conducted under general laboratory conditions acceptable to the Government. Unless otherwise specified herein, all tests shall be conducted at ambient room air temperature. The ambient room temperature shall be maintained at 77° ± 15°F. Unless otherwise specified herein, or in applicable specifications, test specimens shall be thermally stabilized for one hour prior to being subjected to tests.

4.5.1.2 Apparatus. In addition to standard measuring equipment (see table VI and applicable test specification), test apparatus shall include chambers for conducting low temperature (-65°F) and high temperature (160°F) tests, and a submersion tank. A coil having an inductance of 0.20 millihenries shall be included as a series component with the lamp load in all tests.

4.5.1.3 Calibration of test equipment. Test equipment accuracy shall be such as to permit measurement of 10 percent of product or test specification tolerance. Standard measuring instrument tolerance accuracies are listed in table VI.

TABLE VI. Measuring instrument tolerances.

Measuring instrument	Accuracy
Voltmeter dc	± 0.25 percent
Ammeter dc (Multi-range)	± 0.50 percent

4.5.2 Electrical.

4.5.2.1 Flashing. To determine conformance to 3.4.4, the flasher unit shall be subjected to test potentials of 18, 24, and 30 volts dc. The “turn off time” shall be determined at the specified test voltages.

4.5.2.2 Overload.

4.5.2.2.1 Control unit. To determine conformance to 3.4.5.1 the control unit shall be connected into a circuit with a switch, a 30 Amp circuit breaker conforming to MIL-C-13516/1-4 (30 Amp) and a 24 volt power source. The power source shall consist of two fully charged batteries conforming to MS35000, Type 6TN, connected in series or an equivalent regulated power supply. With the switch in the “OFF” position, connect the positive battery

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terminal to the switch, connect the switch to the circuit breaker and connect the circuit breaker to terminal "D" of the control unit (unit in neutral position). Connect terminals "C" and "E" of the control unit to the negative terminal of the power source. The following test shall be performed:

a. With the control unit in the left turn position, turn the test circuit switch to the "ON" position for one minute and then return to the "OFF" position for three minutes.
b. Repeat the cycle five times.
c. With the control unit in the right turn position repeat the above operations.
d. Subsequent to the above test, the flasher and control arm shall meet the requirements of the flashing test.

4.5.2.2.2 Flasher unit. To determine conformance to 3.4.5.2, the flasher unit shall be connected in a test circuit with one A-A-52463-2-10 lamp, one circuit breaker conforming to MIL-C-13516/1-4 (30 amp), necessary wiring, and a power source using two type 6TN batteries connected in series. The batteries shall be fully charged in good condition, with the electrolyte having a corrected specific gravity reading of 1.280 ± 0.005 at 77°F. The circuit breaker shall be connected between the power source and the flasher. The lamp shall be shorted with a 10 inch length of No.14 AWG wire for one minute. After removing the short, the flasher unit shall be tested as specified in 4.5.2.1.

4.5.2.3 Reverse polarity. To determine conformance to 3.4.6, reverse polarity shall be applied to the flasher unit, using a circuit consisting of 10 A-A-52463-2-10 lamps, necessary wiring, and 28 volt dc power source. The reversed polarity shall be applied ten consecutive times for one minute, at two minute intervals. The flasher shall subsequently be tested as specified in 4.5.2.1.

4.5.2.4 Endurance. To determine conformance to 3.4.7, the control unit shall be connected in the test circuit and cycled for 50,000 cycles as specified in 4.5.4.1.1, except the temperature shall be 77° ± 15°F. The control unit shall then be cycled for 123,000 cycles in the same manner, except four A-A-52463-2-10 lamps shall be removed from the load circuit, simulating removal of two lamps from the right rear position and two lamps from the left rear position. The flasher unit shall be cycled for 300 hours as specified in 4.5.4.1.2, except the temperature shall be 77° ± 15°F. The endurance test shall be repeated on the hazard warning switch, except that the switch shall be cycled 5000 times and the cycling rate shall be two seconds "on" and two seconds "off". A cycle for the hazard warning switch shall consist of the lever movement from neutral to hazard warning position and return to neutral.

4.5.2.5 Transient capability. To determine conformance to 3.4.8, the flasher shall be subjected to a properly polarized transient voltage of 80 volts dc for 50 ± 5 milliseconds duration. The voltage shall be applied to the input terminals of the flasher unit ten consecutive times at one minute intervals. The flasher unit shall subsequently be tested as specified in 4.5.2.1.

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4.5.2.6 Electromagnetic interference. To determine conformance to 3.4.9, the control and flasher units shall be proved to meet the requirements for electrical/electromechanical equipment (class C3) of MIL-STD-461 when tested as a component sub-assembly per MIL-STD-462.

4.5.3 Mechanical.

4.5.3.1 Actuation. To determine conformance to 3.5.1, the control unit shall be mounted in a vise or other suitable means. With the lever in one extreme position, a force shall be gradually applied within 0.5 inch of the end of the lever in line with the direction of motion of lever until it actuates. A force shall again be gradually applied in the same direction until the lever actuates again. A force shall be applied to the control lever in the opposite direction until the lever is in neutral position, and again until the lever is in the original position. This procedure shall be repeated ten times. The average force necessary to actuate the control lever shall not exceed the limits specified in 3.5.1.

4.5.3.2 Strength, control unit. To determine conformance to 3.5.2, the control unit shall be mounted on a fixture equipped with a vertical bar 1.5 inches in diameter. The control unit shall be attached to the bar to simulate normal mounting on a vehicle steering column. With the lever hazard lock released and the lever in the farthest forward actuating position, a 70 pound force shall be applied gradually in line with the actuation motion of the lever. The force shall be applied for two minutes at right angles to the lever at a point 1.5 inches from the end of the lever. The lever shall then be placed at the opposite end of the actuation range and a 70 pound force applied as before. The lever shall then be returned to the neutral position and the 70 pound force shall be applied in both directions perpendicular to the plane of normal lever actuation.

4.5.4 Environmental.

4.5.4.1 High temperature.

4.5.4.1.1 Control unit. To determine conformance to 3.6.1.1, the control unit shall be connected in a directional signal light circuit consisting of a flasher unit, ten A-A-52463-2-10 lamps, 28 volt dc power source, and the necessary wiring. The control unit shall be stored in an ambient temperature of 160° $\pm$ 5°F for 24 hours and then cycled for 1000 cycles at that temperature. Control unit cycling shall be conducted in the following sequence: 1 $\pm$ 0.1 second in neutral position, 1 $\pm$ 0.1 second in right turn position, 1 $\pm$ 0.1 second (again) in neutral position, and 1 $\pm$ 0.1 second in left turn position (15 cycles per minute). Cycling shall be accomplished by applying alternate force of not less than two pounds or more than four pounds to the end of the control lever in each direction of motion. The control unit shall conform to 3.5.1 during the test.

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4.5.4.1.2 Flasher unit. To determine conformance to 3.6.1.2, the flasher unit shall be connected in the circuit specified in 4.5.4.1.1. The flasher unit shall be stored in an ambient temperature of 160° ± 5°F for 24 hours and then cycled for 25 hours at that temperature. Cycling shall consist of 15 ± 1 seconds "on" and 15 ± 1 seconds "off" while making and breaking the 10 ampere lamp load. The flasher shall conform to 3.4.4 during the test.

4.5.4.2 Low temperature.

4.5.4.2.1 Control unit. To determine conformance to 3.6.2.1, the control unit shall be subjected to the test specified in 4.5.4.1.1, except that the ambient temperature shall be -65° ± 5°F. The control unit shall conform to 3.5.1 during the test.

4.5.4.2.2 Flasher unit. To determine conformance to 3.6.2.2, the flasher unit shall be subjected to the test specified in 4.5.4.1.2, except that the ambient temperature shall be -65° ± 5°F. The flasher unit shall conform to 3.4.4 during the test.

4.5.4.3 Shock. To determine conformance to 3.6.3, the control and flasher units shall be subjected to the shock test specified in method 202 of MIL-STD-202. Control and flasher units shall be mounted as in intended use. No load shall be connected to units during the test. A force of 50 gravity units shall be applied to the flasher unit once in the direction of each of the major axes (a total of three times). Fifty gravity units shall be applied to the control unit once in the direction of each of the three major axes, with the signal control lever in the up (right turn) position, and center (off) position, and the down (left turn) position (a total of nine times). Subsequent to the test, the control unit shall be tested as specified in 4.5.3.1, and the flasher unit shall be tested as specified in 4.5.2.1.

4.5.4.4 Vibration. To determine conformance to 3.6.4, the control and flasher units shall be subjected to the vibration test specified in method 201 of MIL-STD-202. Control unit shall be securely mounted to a metal cylinder having a diameter within limits specified on 11613632 for the mounting clamp. The axis of the body of the control unit shall be perpendicular to the mounting clamp. The axis of the body of the control unit shall be perpendicular to the vibration table surface. Flasher unit shall be mounted as in intended operation. Vibration shall be applied to the control unit for 20 minutes in the direction of each of the major axes with the control lever in each position (a total of three hours of vibration). During vibration, control units shall be observed for oscillation of the control unit body about the cylinder axis, or displacement of the control lever. Vibration shall be applied to the flasher unit for one hour in the direction of each of the three major axes. Subsequently, the control unit shall be tested as specified in 4.5.3.1 and the flasher unit shall be tested as specified in 4.5.2.1.

4.5.4.5 Waterproofness. To determine conformance to 3.6.5, the control unit and flasher unit shall be subjected to the waterproofness test specified for type I, class 2 components of

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MIL-STD-1184. Subsequently, the control unit shall be tested as specified in 4.5.3.1 and the flasher unit shall be tested as specified in 4.5.2.1.

4.5.4.6 Corrosion resistance. To determine conformance to 3.6.6, the control unit and flasher unit shall be subjected to the salt spray test specified in method 101B of MIL-STD-202 except that the length of the test will be 200 hours. A five percent salt solution shall be used. Subsequently, the control unit shall be tested as specified in 4.5.3.1 and flasher unit shall be tested as specified in 4.5.2.1.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The control and flasher units covered by this specification are intended for use in directional control signal systems and hazard warning systems of military vehicles. This system should not be used for ambient temperatures above 165°F. This specification should not be used for procurement of control units for newly designed vehicles. Control units, for newly designed vehicles, should be self-cancelling by steering wheel rotation in accordance with Federal Motor Vehicle Safety Standards.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Unit required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. If first article inspection is required (see 3.1).
e. Applicable drawing (see 3.3).
f. Any special lot size (see 4.3.1.1).

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g. If test conditions should be other than as specified (see 4.5.1).
h. Packaging requirements (see 5.1).

6.3 Definition of component terms.

6.3.1 Control unit. The control unit is the component mounted on the vehicle steering column, which includes the directional control, hazard warning switch, flash indicator lamp, and connector necessary for electrical connection to the vehicle electrical system. The control unit also includes the switching necessary for supplying positive voltage to the flasher unit when this unit is set in “right hand turn”, “left hand turn”, or “hazard warning” position.

6.3.2 Flasher unit. The flasher unit is the component which, while in operation, makes and breaks the directional signal light circuit at the prescribed rate and has no moving parts.

6.3.3 Hazard warning switch. The hazard warning switch is a driver-controlled device which causes all turn signal lamps to flash simultaneously to indicate the presence of a vehicular hazard to approaching drivers.

6.4 Definition of operation terms.

6.4.1 Flashing. “Flashing” of a directional control system is defined as the variation of the luminosity of each of the signal lamp bulbs from less than that noticeable to a person with normal or corrected 20/20 vision to a luminosity greater than 25 candle power to the original level.

6.4.2 On time. “On time” is defined as the time elapsed between the period when full voltage (18 to 30 volts dc) is applied to the load and full voltage is switched from the load.

6.4.3 Off time. “Off time” is defined as the time elapsed between the period when full voltage (18 to 30 volts dc) is switched from the load and full voltage is applied to the load.

6.4.4 Flashing rate. The “flashing rate” in flashes per minute is defined as follows:

$$\text{Flashing rate} = \frac{60}{\text{on time + off time (expressed in seconds)}}$$

Where computation is carried to the nearest tenth and “on time” and “off time” are consecutive and as defined in 6.4.2 and 6.4.3, respectively.

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6.4.5 Percent on time. "Percent on time" is defined as follows:

$$\text{Percent on time} = \frac{\text{on time}}{\text{on time} \pm \text{off time (expressed in seconds)}} \times 100$$

Where computation is to be carried to the nearest tenth, and "on time" and "off time" are consecutive and as defined in 6.4.2 and 6.4.3, respectively.

6.4.6 Turn on (starting) time. "Turn on time" is defined as the time elapsed between an instant that the control lever is moved from neutral position into right hand (RH) or left hand (LH) position and the instant that the total current to the RH or LH tail lamps, as applicable, reaches two amperes.

6.4.7 Turn off time. "Turn off time" is defined as the time elapsed between an instant that the control lever is moved from RH or LH position into neutral position and the instant that all noticeable lamp luminosity ceases.

6.4.8 Subject term (key word) listing.

Electrical
Flasher
Hazard warning
Vehicle

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 2590-0243)

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